

AKI — SCENE AKI-02: The Poison Barrel Dock



1. Myoglobin barrel

WHAT YOU SEE

An angry, flexing muscle-bound barrel labeled HIGH CK / MYOGLOBIN, shackled with chains at its base — the bulging muscle bursting open literally IS skeletal-muscle breakdown spilling myoglobin and CK into the blood.

WHAT IT ENCODES

Rhabdomyolysis (crush injury, prolonged immobilization, seizures, statins/fibrates, cocaine, extreme exertion, NMS/malignant hyperthermia) breaks down skeletal muscle, releasing myoglobin and markedly elevating CK — usually >5,000 and AKI risk climbs sharply above ~15,000–20,000 U/L. Associated labs: hyperkalemia, hyperphosphatemia, HYPOcalcemia early, and a later rebound hypercalcemia. Myoglobin is a heme pigment that is filtered and damages tubules.

BOARD TRAP

Diagnosing rhabdomyolysis off a positive blood dipstick alone — the discriminator is a high CK with FEW RBCs on micro; a normal CK essentially excludes it, and the pigment, not RBCs, drives the dipstick reaction.

2. Rhabdo mechanisms

WHAT YOU SEE

Vasoconstriction clamp, direct toxicity, cast obstruction, acidic urine worse

WHAT IT ENCODES

Myoglobin causes AKI by three mechanisms: (1) renal vasoconstriction reducing medullary blood flow, (2) direct heme-mediated proximal tubular toxicity via oxidative injury, and (3) intratubular cast obstruction when myoglobin binds Tamm-Horsfall protein. ACIDIC urine promotes cast precipitation and amplifies toxicity, which is the rationale for urine alkalization.

BOARD TRAP

Forgetting that aciduria worsens pigment cast formation — students recall 'give fluids' but miss that acidic urine accelerates injury, which is why alkalinizing the urine (raising pH) is considered to keep myoglobin soluble.

3. Rhabdo treatment

WHAT YOU SEE

Aggressive fluids + alkalinize bag

WHAT IT ENCODES

The cornerstone of rhabdomyolysis management is early, aggressive IV isotonic fluid resuscitation (often several liters, titrated to a urine output of ~200–300 mL/h) to dilute and flush pigment and casts and restore perfusion. Sodium bicarbonate to alkalinize urine (target pH >6.5) is considered to reduce cast formation, though evidence is debated. Treat hyperkalemia urgently; avoid calcium unless symptomatic given rebound hypercalcemia.

BOARD TRAP

Reaching for mannitol or bicarbonate before adequate volume repletion — aggressive crystalloid is the proven first-line; mannitol/bicarbonate are adjuncts, and giving calcium for asymptomatic early hypocalcemia risks precipitating calcium-phosphate and rebound hypercalcemia.

4. Hemoglobin barrel

WHAT YOU SEE

Pink HEMOGLOBIN barrel, transfusion rxn / mechanical valves

WHAT IT ENCODES

Massive intravascular hemolysis releases free hemoglobin that, once haptoglobin is saturated, is filtered and causes pigment nephropathy by the same mechanisms as myoglobin. Causes include ABO-incompatible transfusion reactions, mechanical heart valves, paroxysmal nocturnal hemoglobinuria, severe malaria (blackwater fever), and G6PD-deficiency oxidative crises. Plasma appears pink-red in hemoglobinemia.

BOARD TRAP

Confusing hemoglobinuria with myoglobinuria — both give a blood-positive dipstick with few RBCs; the tell is pink plasma and low haptoglobin/high LDH in hemolysis versus clear plasma and very high CK in rhabdomyolysis.

5. Low haptoglobin

WHAT YOU SEE

Ghost LOW HAPTO

WHAT IT ENCODES

Haptoglobin binds free hemoglobin for clearance, so a LOW haptoglobin is a sensitive marker of intravascular hemolysis. The full hemolysis panel shows low haptoglobin, elevated LDH, elevated indirect (unconjugated) bilirubin, and reticulocytosis; schistocytes on smear indicate a microangiopathic/mechanical cause (TMA, mechanical valve).

BOARD TRAP

Using low haptoglobin to claim rhabdomyolysis — haptoglobin binds hemoglobin, not myoglobin, so it is a hemolysis marker; in pure rhabdo, haptoglobin is normal and CK is the discriminating test.

6. Dipstick + few RBCs

WHAT YOU SEE

DIPSTICK+ FEW RBCs = pigment, not hematuria card

WHAT IT ENCODES

Pigment nephropathy fingerprint: the urine dipstick is POSITIVE for blood (the heme moiety of myoglobin/hemoglobin triggers the peroxidase pad) but microscopy shows FEW or NO red blood cells. This dipstick-positive/RBC-negative dissociation distinguishes pigment from true hematuria (where intact RBCs are seen).

BOARD TRAP

Reading a blood-positive dipstick as confirmed hematuria and launching a urologic bleed workup — with few RBCs on micro, the answer is pigmenturia (myoglobin or hemoglobin), confirmed by CK and a hemolysis panel rather than cystoscopy.

7. Dipstick negative / SSA

WHAT YOU SEE

DIPSTICK NEGATIVE no protein - use SSA card

WHAT IT ENCODES

The urine dipstick protein pad detects ALBUMIN, not immunoglobulin light chains, so a patient with heavy Bence Jones (light-chain) proteinuria can have a NEGATIVE dipstick. The sulfosalicylic acid (SSA) test precipitates ALL urine proteins including light chains, so a dipstick-negative but SSA-positive urine reveals non-albumin protein — the classic myeloma clue. Confirm with urine protein electrophoresis/immunofixation.

BOARD TRAP

Concluding 'no proteinuria' from a negative dipstick in a patient with bone pain, anemia, and hypercalcemia — the dipstick misses light chains; the discriminator is a positive SSA test or UPEP, and missing it overlooks myeloma cast nephropathy.

8. Light chains / myeloma

WHAT YOU SEE

LIGHT CHAINS FRAGILE crate, MULTIPLE MYELOMA, SSA test

WHAT IT ENCODES

In multiple myeloma, overproduced monoclonal free light chains are filtered and form obstructing casts with Tamm-Horsfall protein (cast nephropathy), the leading cause of myeloma AKI. The diagnostic combination is dipstick-negative proteinuria but SSA-positive, plus serum/urine free light chains and an M-spike on electrophoresis. Suspect with the CRAB features: hyperCalcemia, Renal failure, Anemia, Bone lesions.

BOARD TRAP

Attributing myeloma AKI purely to hypercalcemia and stopping there — cast nephropathy from light chains is the dominant and often irreversible mechanism, so treatment targets the plasma-cell clone and reduces light-chain load, not just the calcium.

9. Myeloma mechanisms

WHAT YOU SEE

Cast nephropathy, direct toxicity, hypercalcemia vasoconstriction

WHAT IT ENCODES

Myeloma injures the kidney through multiple pathways: (1) cast nephropathy from light chains obstructing distal tubules, (2) direct proximal tubular toxicity from reabsorbed light chains (can cause acquired Fanconi syndrome), and (3) hypercalcemia-induced afferent vasoconstriction and volume depletion (nephrogenic DI). Amyloidosis (AL) and light-chain deposition disease add chronic glomerular damage.

BOARD TRAP

Giving IV contrast or NSAIDs to a dehydrated myeloma patient — these precipitate cast nephropathy; the right move is aggressive hydration, avoiding nephrotoxins, correcting hypercalcemia, and urgently lowering the light-chain burden.

10. Uric acid / TLS

WHAT YOU SEE

URIC ACID ice crystal block, TUMOR LYSIS, 'K up P up U up Ca down'

WHAT IT ENCODES

Tumor lysis syndrome follows cytotoxic therapy (or spontaneously) in bulky, high-turnover malignancies (Burkitt lymphoma, ALL, high-grade lymphomas), dumping intracellular contents into the blood. The metabolic signature is HYPERkalemia, HYPERphosphatemia, HYPERuricemia, and HYPOcalcemia (phosphate binds calcium). Uric acid and calcium-phosphate crystals precipitate in tubules causing obstructive AKI.

BOARD TRAP

Replacing calcium for the hypocalcemia of TLS — giving calcium in the face of hyperphosphatemia precipitates calcium-phosphate in tissues and tubules, worsening injury; only treat hypocalcemia if symptomatic (tetany, arrhythmia).

11. Crystal obstruction

WHAT YOU SEE

CRYSTAL OBSTRUCTION clock at bottom

WHAT IT ENCODES

TLS-related AKI is intratubular crystal obstruction by uric acid (worse in acidic urine) and calcium-phosphate (worse in alkaline urine). Prevention/treatment: aggressive IV hydration to maintain high urine flow; rasburicase (recombinant urate oxidase) for high-risk or established TLS to enzymatically degrade uric acid; allopurinol or febuxostat to block new urate formation in lower-risk patients.

BOARD TRAP

Routinely alkalinizing urine in TLS as you would for rhabdo — alkalinization helps uric acid solubility but promotes calcium-phosphate precipitation, so it is no longer recommended; the answer is hydration plus rasburicase, not bicarbonate.

12. Ethylene glycol

WHAT YOU SEE

Green ETHYLENE GLYCOL jug, HIGH ANION GAP MUDPILE

WHAT IT ENCODES

Ethylene glycol (antifreeze) is metabolized by alcohol dehydrogenase to glycolic and oxalic acids, producing a HIGH anion-gap metabolic acidosis (an 'M' in MUDPILES) with an elevated OSMOLAR gap early. Oxalate binds calcium forming calcium-oxalate crystals (envelope/needle-shaped) that cause AKI and hypocalcemia; urine may fluoresce under Wood's lamp. Methanol, by contrast, causes visual loss/optic injury.

BOARD TRAP

Confusing ethylene glycol with methanol — both give a high anion-gap acidosis with an osmolar gap and are treated with fomepizole, but ethylene glycol causes AKI and calcium-oxalate crystalluria/hypocalcemia, whereas methanol causes blindness and optic disc hyperemia.

13. Toxic alcohol Rx

WHAT YOU SEE

Fomepizole / IV alcohol bag, HD if severe, '>15 mg/dL'

WHAT IT ENCODES

Treat toxic-alcohol (ethylene glycol or methanol) poisoning by blocking alcohol dehydrogenase with FOMEPIZOLE (preferred) or ethanol infusion to stop formation of toxic metabolites. Add hemodialysis for severe metabolic acidosis, end-organ injury (AKI, visual changes), or high measured levels (often cited around >50 mg/dL, with treatment thresholds and the depicted >15–20 range varying by source); give cofactors — thiamine/pyridoxine for ethylene glycol, folate/folinic acid for methanol.

BOARD TRAP

Withholding fomepizole until a confirmatory level returns — empiric fomepizole is started on clinical suspicion (high anion-gap and osmolar gap, history) because waiting allows ongoing toxic-metabolite production and irreversible AKI or blindness.
